

Mathematical Analysis of Quantum Electromagnetic Coupling

Based on QED Decoupling at the Event Horizon

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September 28, 2026

Abstract

The paper proposes a new mathematical model that explains the phenomenon of "event horizon darkness" ($r = r_s$) without the need for the concept of a "light-trapping mechanical gravitational force". The paper mathematically proves that the absence of radiation emitted from the event horizon is not due to the trapping of photons—being massless quantum excitations—inside a gravitational cage, but rather the result of quantum electrodynamic (QED) decoupling caused by the vanishing density of material electron charges ($N_e \rightarrow 0$) at the boundary (Decoupling). The quantum state of the electromagnetic field at the horizon shifts from a classically interacting field into a pure quantum vacuum state $\langle 0 | \hat{T}_{\mu\nu} | 0 \rangle$, making the "light trapping" phenomenon merely an optical/perceptual illusion resulting from the absence of electromagnetic interaction mediators.

1 Mathematical Derivations and Quantum Mathematics

1.1 A. QED Decoupling Dynamics and Current Vanishing

We begin with the Lagrangian density of Quantum Electrodynamics (QED) theory in curved Schwarzschild spacetime:

$$\mathcal{L}_{QED} = -g \left(\bar{\psi} (i\gamma^\mu D_\mu - m_e) \psi - \frac{1}{4} F_{\mu\nu} F^{\mu\nu} \right) \quad (1)$$

where A_μ is the photon vector potential, $D_\mu = \partial_\mu + ieA_\mu$ is the covariant derivative, and ψ is the electron-positron field.

The electromagnetic interaction term responsible for the appearance, emission, and scattering of light is given by:

$$\mathcal{L}_{int} = -g \cdot J^\mu A_\mu \quad (2)$$

where the electric charge current is given by $J^\mu = e\bar{\psi}\gamma^\mu\psi$.

When approaching the event horizon ($r \rightarrow r_s$), electrons undergo a critical discharge resulting from thermal evaporation and the Schwinger mechanism, yielding an exponentially decreasing electron density $N_e(r)$:

$$N_e(r) = N_0 \exp\left(-\lambda \frac{r_s}{r - r_s}\right) \quad (3)$$

Right at the event horizon ($r = r_s$):

$$\lim_{r \rightarrow r_s^+} N_e(r) = 0 \implies \lim_{r \rightarrow r_s^+} \langle J^\mu \rangle = 0 \quad (4)$$

Since the charge current approaches zero, the interaction term vanishes completely:

$$\lim_{r \rightarrow r_s^+} \mathcal{L}_{int} = 0 \quad (5)$$

First Mathematical Result: The vanishing of the interaction term ($\mathcal{L}_{int} = 0$) implies that remaining photons cannot be reflected, scattered, or emitted, as the electromagnetic field decouples completely from matter.

1.2 B. Transition of the Energy-Momentum Tensor to Quantum Vacuum

The classical energy-momentum tensor of the electromagnetic field is given by:

$$T_{\mu\nu}^{(EM)} = F_{\mu\alpha} F_\nu^\alpha - \frac{1}{4} g_{\mu\nu} F_{\alpha\beta} F^{\alpha\beta} \quad (6)$$

Due to decoupling ($J^\mu \rightarrow 0$) and the absence of electrons at $r = r_s$, the classical expectation value of the field vanishes ($\langle F_{\mu\nu} \rangle = 0$), and the state transitions to the Quantum Vacuum Expectation Value:

$$\lim_{r \rightarrow r_s^+} \langle T_{\mu\nu} \rangle = \langle 0 | \hat{T}_{\mu\nu}^{(vacuum)} | 0 \rangle_{Renorm} \quad (7)$$

Using the Boulware Vacuum State (or Hartle-Hawking) at the event horizon, we find that the temporal component of the energy-momentum tensor records pure vacuum fluctuations:

$$\langle \hat{T}_t^t \rangle_{vacuum} \sim -\frac{\hbar c}{768\pi^2 r_s^4 \left(1 - \frac{r_s}{r}\right)} \quad (8)$$

This equation proves that what exists at the event horizon is not trapped light matter, but rather quantum vacuum stress that yields evaporative Hawking radiation at the boundary.

1.3 C. Observed Intensity Equation

To calculate the total optical intensity I_{obs} received by a distant observer ($r \rightarrow \infty$) we integrate the effect of gravitational redshift with the electron decay function:

$$I_{obs}(r) = I_0 \cdot \left(1 - \frac{r_s}{r}\right)^2 \cdot \frac{N_e(r)}{N_0} \quad (9)$$

At the event horizon ($r \rightarrow r_s^+$):

$$\lim_{r \rightarrow r_s^+} I_{obs}(r) = I_0 \cdot (0) \cdot (0) = 0 \quad (10)$$

1.4 D. Quantum-Geometric Transition Amplitude and Null Geodesics

To rigorously reconcile the QED decoupling mechanism with the null geodesics of General Relativity, we formulate the overall quantum-geometric transition amplitude $\mathcal{A}(r_s \rightarrow \infty)$ for an electromagnetic signal originating near the horizon boundary:

$$\mathcal{A}(r_s \rightarrow \infty) = \int d^4x \sqrt{-g} \left\langle 0 \left| \hat{J}^\mu(r_\infty) \hat{A}_\mu(r_\infty) \cdot \mathcal{G}_{\mu\nu}^{(\text{ret})}(x_\infty, x_s) \cdot \hat{J}^\nu(r_s) \hat{A}^\nu(r_s) \right| 0 \right\rangle \quad (11)$$

where $\mathcal{G}_{\mu\nu}^{(\text{ret})}(x_\infty, x_s)$ denotes the retarded photon propagator representing the classical null geodesic trajectory in curved Schwarzschild spacetime, and $\hat{J}^\nu(r_s)$ is the quantum charge current density operator at the boundary.

Taking the explicit decoupling limit as specified in Eq. (4), where the material electron charge density vanishes at the boundary ($N_e(r) \rightarrow 0$ as $r \rightarrow r_s^+$):

$$\lim_{r \rightarrow r_s^+} \hat{J}^\nu(r) = 0 \implies \lim_{r \rightarrow r_s^+} \mathcal{A}(r_s \rightarrow \infty) = 0 \quad (12)$$

This fundamental result demonstrates that while the geometric path operator remains non-zero ($\mathcal{G}_{\mu\nu}^{(\text{ret})} \neq 0$), the physical transition probability vanishes identically due to the zero-current boundary condition. Consequently, the spacetime trajectory becomes an unpopulated state, proving that event horizon darkness stems from quantum source extinction rather than mechanical confinement.

The zero obtained here is twofold:

1. **Geometric zero:** Due to the asymptotic infinite redshift.
2. **Quantum zero:** Due to the absence of charged matter ($N_e = 0$) that interacts and generates optical signals.

2 Physical Discussion

The mathematical derivations above confirm the validity of the hypothesis:

1. Light is a quantum entity with zero rest mass ($m_0 = 0$); it is impossible to formulate a Newtonian "pulling" or mechanical "trapping" equation for the photon because the photon is an excitation in an electromagnetic field, not a material body.
2. **Conceptual Illusion:** The conventional explanation isolates only the geometric part and concludes that light is trapped inside. However, the quantum formulation proves that the optical phenomenon itself has vanished and been invalidated due to the lack of electrons to interact with it; what arises at the horizon are free quantum vacuum fluctuations.

3 Conclusion

The paper concludes that the event horizon is not a prison for light, but rather a boundary where material electromagnetic interaction terminates and quantum vacuum fluctuations dominate the geometric scene. This renders the description of "light trapping" an apparent perceptual characterization rather than a mechanical-kinematic reality.

References

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